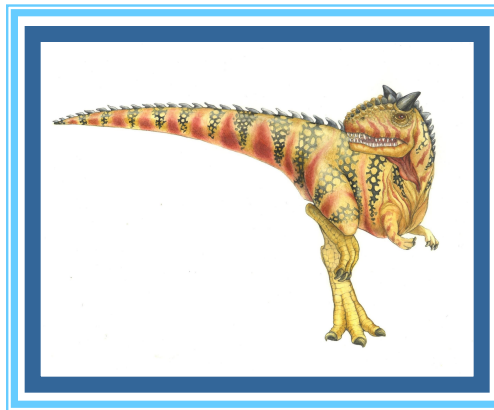


Chapter 3: Processes





Outline

- Process Concept
- Process Scheduling
- Operations on Processes
- Interprocess Communication
- IPC in Shared-Memory Systems
- IPC in Message-Passing Systems
- Examples of IPC Systems
- Communication in Client-Server Systems





Objectives

- Identify the separate components of a process and illustrate how they are represented and scheduled in an operating system.
- Describe how processes are created and terminated in an operating system, including developing programs using the appropriate system calls that perform these operations.
- Describe and contrast interprocess communication using shared memory and message passing.
- Design programs that uses pipes and POSIX shared memory to perform interprocess communication.
- Describe client-server communication using sockets and remote procedure calls.
- Design kernel modules that interact with the Linux operating system.





Process Concept

- An operating system executes a variety of programs that run as a process.
- **Process** – a program in execution; process execution must progress in sequential fashion. No parallel execution of instructions of a single process
- Multiple parts
 - The program code, also called **text section**
 - Current activity including **program counter**, processor registers
 - **Stack** containing temporary data
 - 4 Function parameters, return addresses, local variables
 - **Data section** containing global variables
 - **Heap** containing memory dynamically allocated during run time





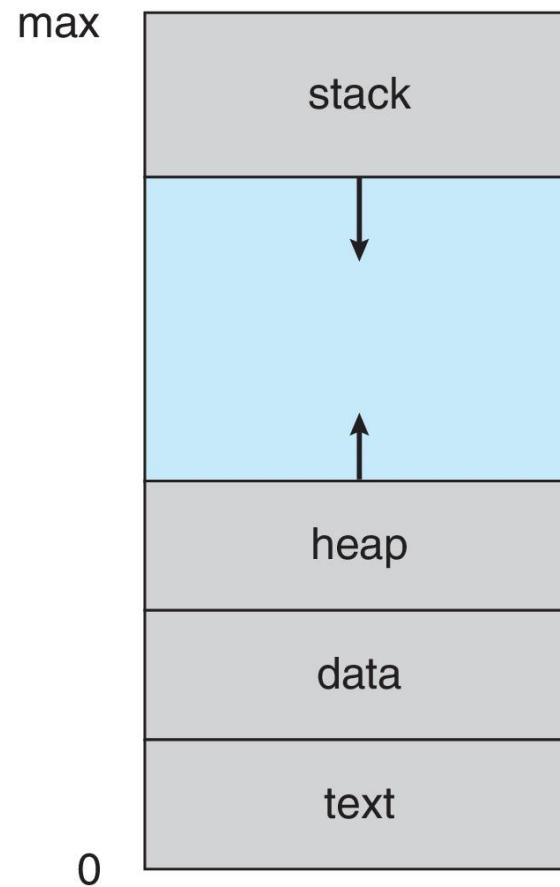
Process Concept (Cont.)

- Program is **passive** entity stored on disk (**executable file**); process is **active**
 - Program becomes process when an executable file is loaded into memory
- Execution of program started via GUI mouse clicks, command line entry of its name, etc.
- One program can be several processes
 - Consider multiple users executing the same program



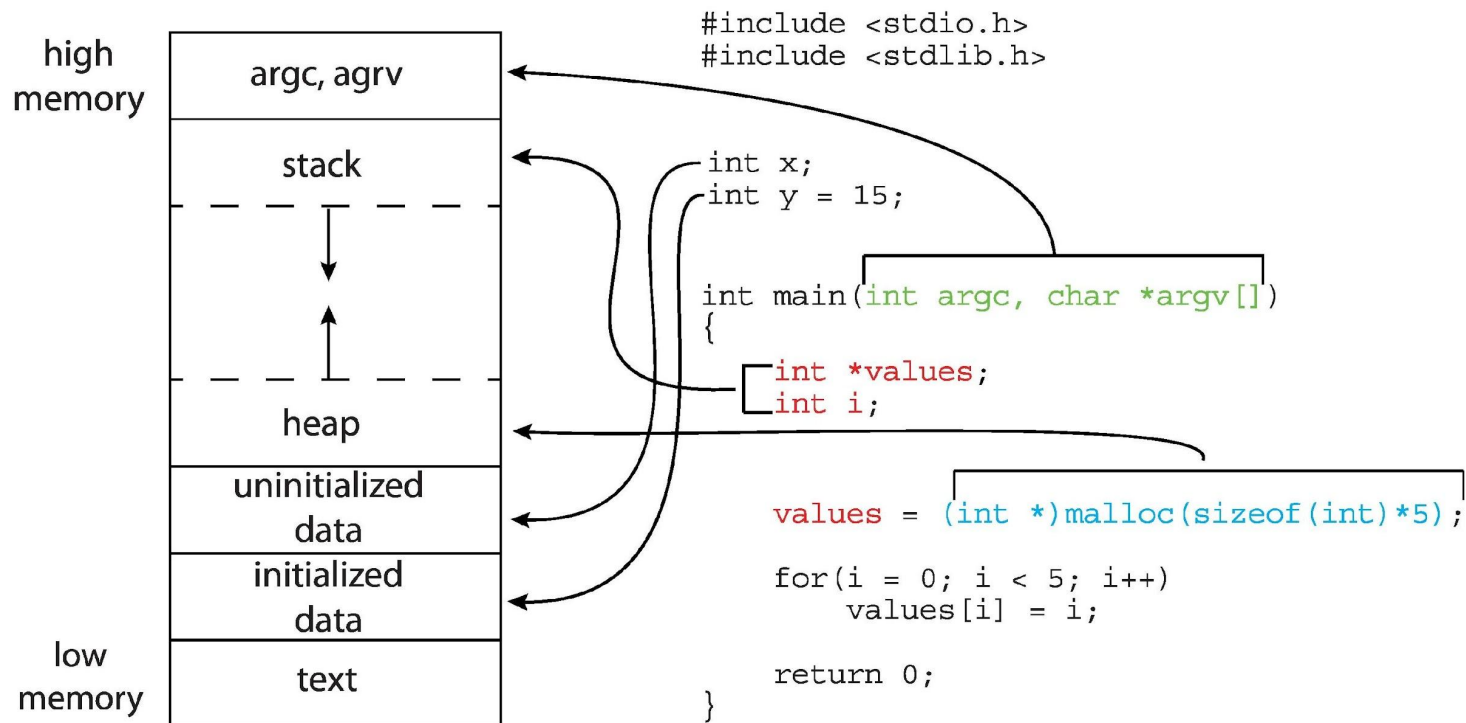


Process in Memory





Memory Layout of a C Program





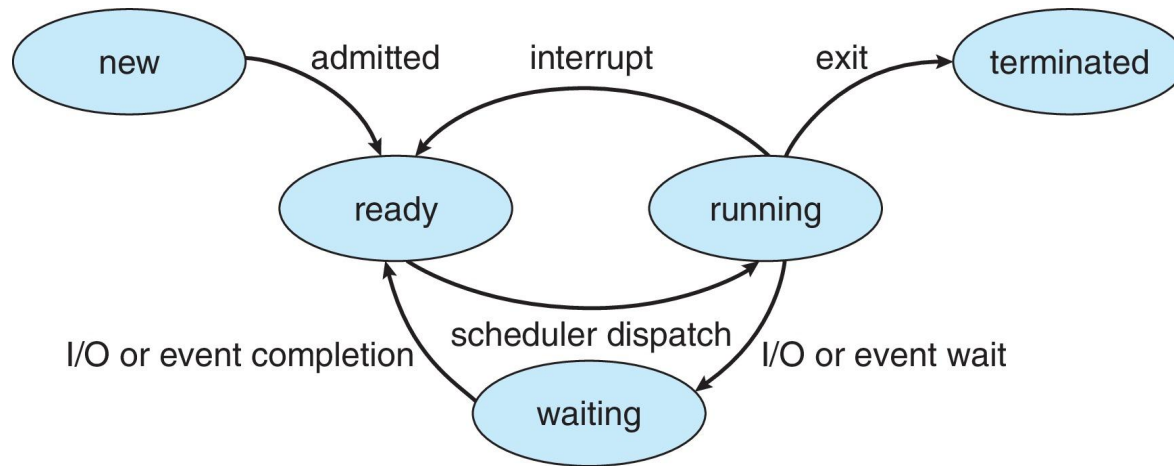
Process State

- As a process executes, it changes **state**
 - **New:** The process is being created
 - **Running:** Instructions are being executed
 - **Waiting:** The process is waiting for some event to occur
 - **Ready:** The process is waiting to be assigned to a processor
 - **Terminated:** The process has finished execution





Diagram of Process State





Process Control Block (PCB)

Information associated with each process(also called **task control block**)

- Process state – running, waiting, etc.
- Program counter – location of instruction to next execute
- CPU registers – contents of all process-centric registers
- CPU scheduling information- priorities, scheduling queue pointers
- Memory-management information – memory allocated to the process
- Accounting information – CPU used, clock time elapsed since start, time limits
- I/O status information – I/O devices allocated to process, list of open files

process state
process number
program counter
registers
memory limits
list of open files
...





Threads

- So far, process has a single thread of execution
- Consider having multiple program counters per process
 - Multiple locations can execute at once
 - 4 Multiple threads of control -> **threads**
- Must then have storage for thread details, multiple program counters in PCB
- Explore in detail in Chapter 4

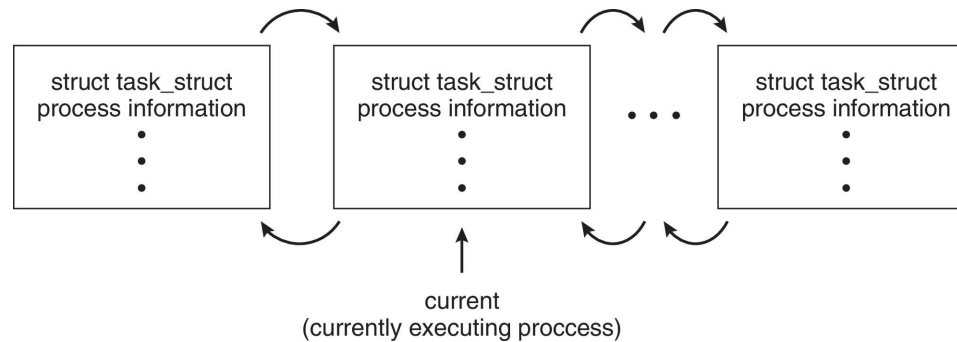




Process Representation in Linux

Represented by the C structure `task_struct`

```
pid t_pid;           /* process identifier */
long state;          /* state of the process */
unsigned int time_slice /* scheduling information */
struct task_struct *parent; /* this process's parent */
struct list_head children; /* this process's children */
struct files_struct *files; /* list of open files */
struct mm_struct *mm; /* address space of this process */
```





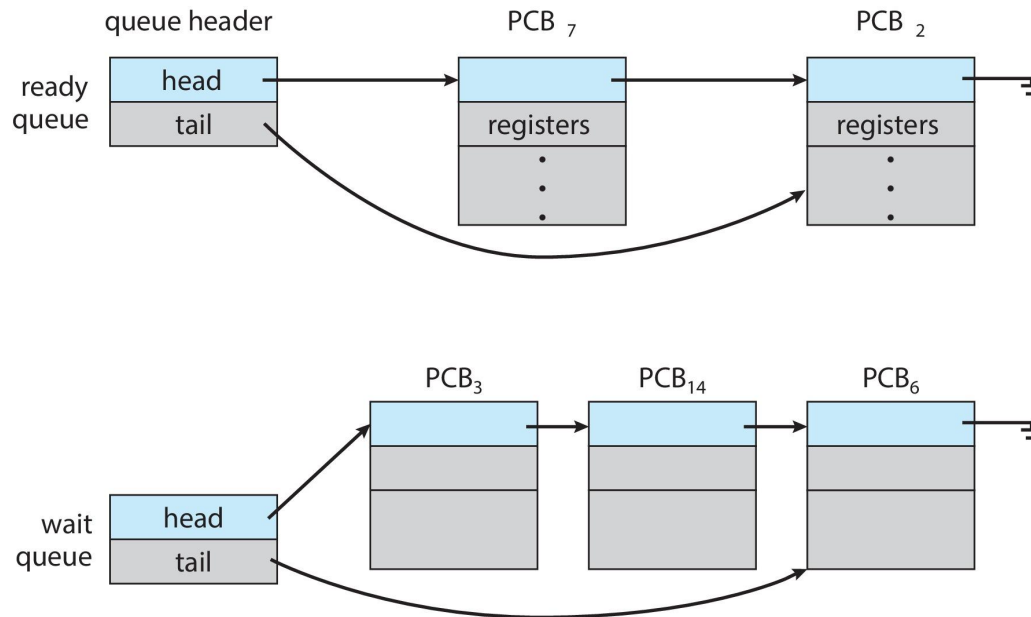
Process Scheduling

- **Process scheduler** selects among available processes for next execution on CPU core
- Goal -- Maximize CPU use, quickly switch processes onto CPU core
- Maintains **scheduling queues** of processes
 - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
 - **Wait queues** – set of processes waiting for an event (i.e., I/O)
 - Processes migrate among the various queues



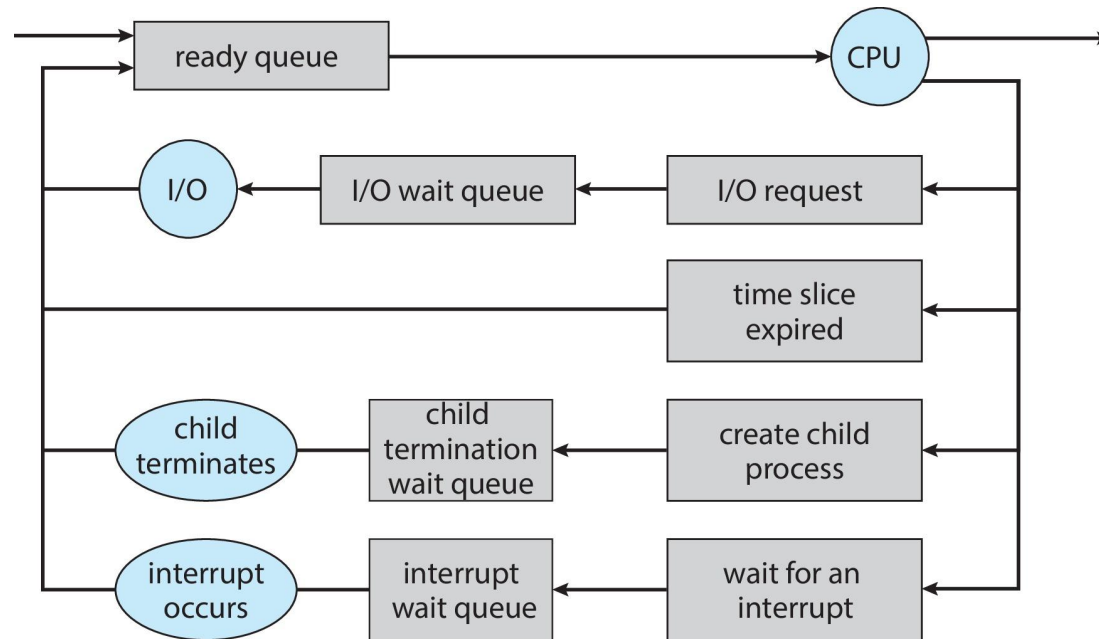


Ready and Wait Queues





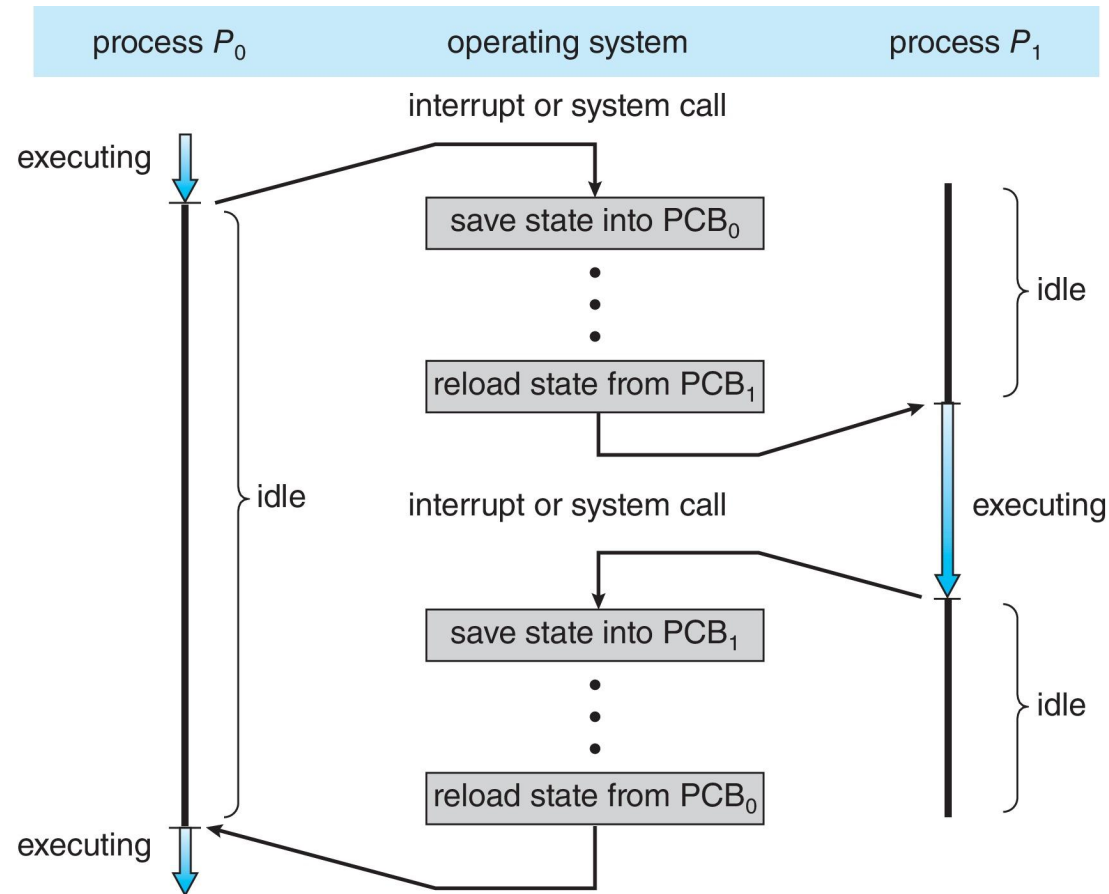
Representation of Process Scheduling





CPU Switch From Process to Process

A **context switch** occurs when the CPU switches from one process to another.





Context Switch

- When CPU switches to another process, the system must **save the state** of the old process and load the **saved state** for the new process via a **context switch**
- **Context** of a process represented in the PCB
- **Context-switch time is pure overhead;** the system does no useful work while switching
 - The more complex the OS and the PCB □ the longer the context switch
- Time dependent on hardware support
 - Some hardware provides multiple sets of registers per CPU □ multiple contexts loaded at once





Multitasking in Mobile Systems

- Some mobile systems (e.g., early version of iOS) allow only one process to run, others suspended
- Due to screen real estate, user interface limits iOS provides for a
 - Single **foreground** process- controlled via user interface
 - Multiple **background** processes– in memory, running, but not on the display, and with limits
 - Limits include single, short task, receiving notification of events, specific long-running tasks like audio playback
- Android runs foreground and background, with fewer limits
 - Background process uses a **service** to perform tasks
 - Service can keep running even if background process is suspended
 - Service has no user interface, small memory use





Operations on Processes

- System must provide mechanisms for:
 - Process creation
 - Process termination





Process Creation

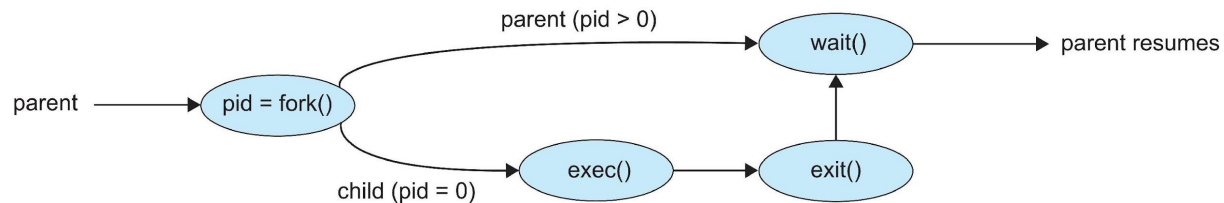
- **Parent** process create **children** processes, which, in turn create other processes, forming a **tree** of processes
- Generally, process identified and managed via a **process identifier (pid)**
- Resource sharing options
 - Parent and children share all resources
 - Children share subset of parent's resources
 - Parent and child share no resources
- Execution options
 - Parent and children execute concurrently
 - Parent waits until children terminate





Process Creation (Cont.)

- Address space
 - Child duplicate of parent
 - Child has a program loaded into it
- UNIX examples
 - **fork()** system call creates new process
 - **exec()** system call used after a **fork()** to replace the process' memory space with a new program
 - Parent process calls **wait()** waiting for the child to terminate





Core Shared Resources

- **File Descriptors:**

- The child inherits **references** to the parent's open files.
- **Key Implication:** They share the same file offset. If the child reads 10 bytes, the parent's file position also advances by 10 bytes.

- **Memory Mappings:**

- Regions created with **mmap** and the **MAP_SHARED** flag are shared.
- **Key Implication:** Writes to this memory by one process are instantly visible to the other.

- **Signal Handlers:**

- The child inherits the parent's functions for handling signals.





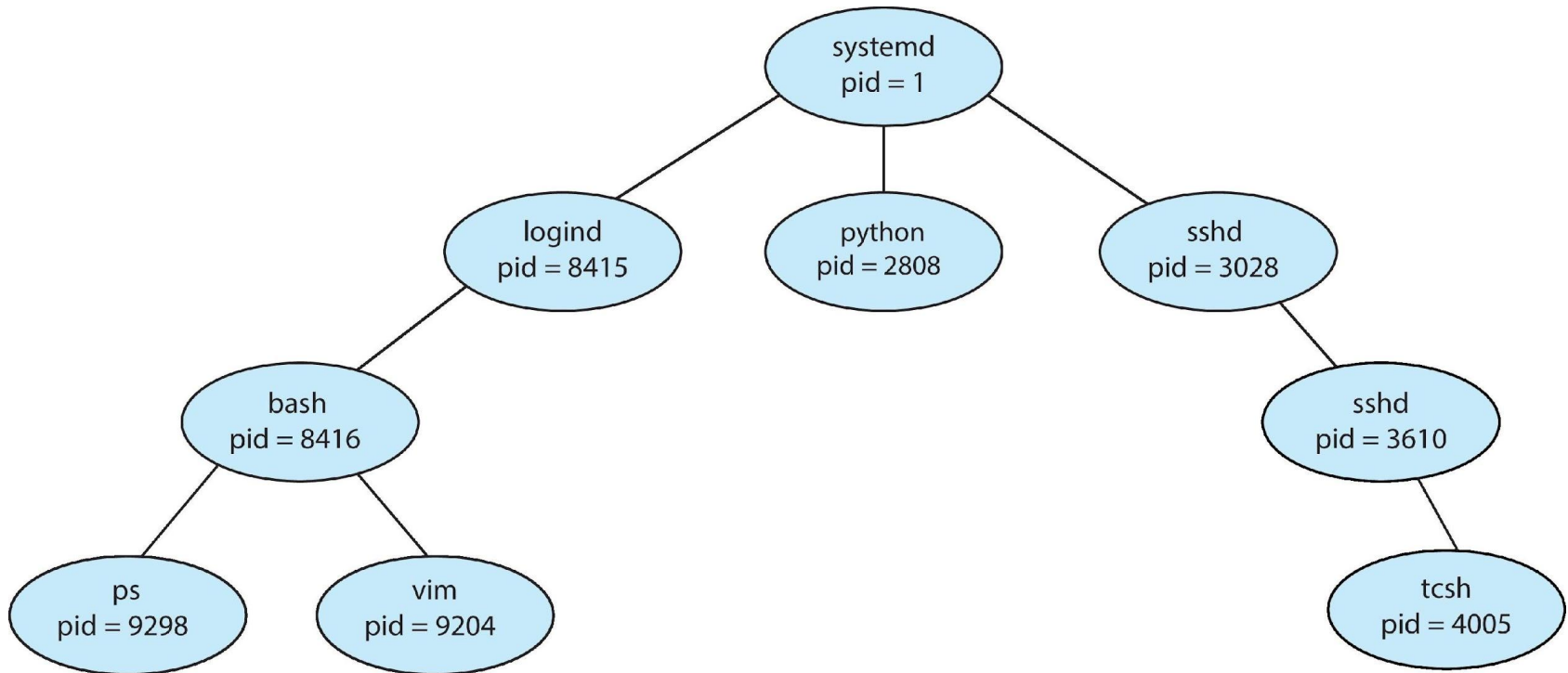
Inherited but Independent (Copy-on-Write)

- These resources are *copied* (or use Copy-on-Write), not shared. Changes in one process do **NOT** affect the other:
- **Memory (Heap/Stack):**
 - The child gets a duplicate of the parent's memory pages.
 - *Note: Modern systems use Copy-on-Write for efficiency, but logically they are separate.*
- **Process Credentials:**
 - Environment variables, umask, and current working directory are duplicated.
- **Open File Descriptors**
 - While the *offset* is shared, the *file descriptor flags* (FD_CLOEXEC) are independent (fd=3)





A Tree of Processes in Linux





C Program Forking Separate Process

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    else if (pid == 0) { /* child process */
        execlp("/bin/ls", "ls", NULL);
    }
    else { /* parent process */
        /* parent will wait for the child to complete */
        wait(NULL);
        printf("Child Complete");
    }

    return 0;
}
```





Creating a Separate Process via Windows API

```
#include <stdio.h>
#include <windows.h>

int main(VOID)
{
    STARTUPINFO si;
    PROCESS_INFORMATION pi;

    /* allocate memory */
    ZeroMemory(&si, sizeof(si));
    si.cb = sizeof(si);
    ZeroMemory(&pi, sizeof(pi));

    /* create child process */
    if (!CreateProcess(NULL, /* use command line */
        "C:\\WINDOWS\\system32\\mspaint.exe", /* command */
        NULL, /* don't inherit process handle */
        NULL, /* don't inherit thread handle */
        FALSE, /* disable handle inheritance */
        0, /* no creation flags */
        NULL, /* use parent's environment block */
        NULL, /* use parent's existing directory */
        &si,
        &pi))
    {
        fprintf(stderr, "Create Process Failed");
        return -1;
    }
    /* parent will wait for the child to complete */
    WaitForSingleObject(pi.hProcess, INFINITE);
    printf("Child Complete");

    /* close handles */
    CloseHandle(pi.hProcess);
    CloseHandle(pi.hThread);
}
```





Process Termination

- Process executes last statement and then asks the operating system to delete it using the **exit()** system call.
 - Returns status data from child to parent (via **wait()**)
 - Process' resources are deallocated by operating system
- Parent may terminate the execution of children processes using the **abort()** system call. Some reasons for doing so:
 - Child has exceeded allocated resources
 - Task assigned to child is no longer required
 - The parent is exiting, and the operating systems does not allow a child to continue if its parent terminates





Process Termination

- Some operating systems do not allow child to exist if its parent has terminated. If a process terminates, then all its children must also be terminated.
 - **cascading termination.** All children, grandchildren, etc., are terminated.
 - The termination is initiated by the operating system.
- The parent process may wait for termination of a child process by using the **wait()** system call. The call returns status information and the pid of the terminated process

```
pid = wait(&status);
```
- If no parent waiting (did not invoke **wait()**) process is a **zombie**
- If parent terminated without invoking **wait()**, process is an **orphan**





Zombie Process

- A **zombie** is a process that has **completed execution** via `exit()` but **still has an entry in the process table**. It is dead but not yet buried.
- **State**: In UNIX/Linux, this is represented as the process status **Z** (defunct).
- It exists to store the **exit status** of the child process (to tell the parent if it succeeded or failed).
- It waits for the parent to call the `wait()` system call to read that status.
- Zombies **do not use memory** (code, stack, heap are freed), but they **do hold a spot** in the kernel's process table.
- If the parent fails to call `wait()` and creates many children, the process table fills up, preventing new processes from being created.
- The parent **must call** `wait()` to "reap" the zombie, removing its entry.
- If the parent dies, orphaned children are adopted by `init` (PID 1), which automatically reaps them.





Orphan Process

- An **orphan** is a process whose **parent has terminated** (died) while the child is still running.
- Parent crashes or exits before the child finishes its job.
- Orphans are **adopted** by the **init** process (PID 1 on Linux/Unix).
- **init** becomes the new parent and automatically calls **wait()** to reap the child when it ends.
- Unlike zombies, orphans **do not** harm the system. They run normally and are cleaned up properly by **init**.
- A child left at a fire station → The state automatically takes over care.





Android Process Importance Hierarchy

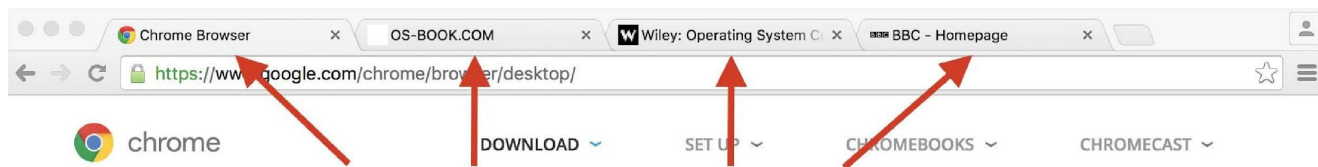
- Mobile operating systems often have to terminate processes to reclaim system resources such as memory. From **most** to **least** important:
 - Foreground process
 - Visible process
 - Service process
 - Background process
 - Empty process
- Android will begin terminating processes that are least important.





Multiprocess Architecture – Chrome Browser

- Many web browsers ran as single process (some still do)
 - If one web site causes trouble, entire browser can hang or crash
- Google Chrome Browser is multiprocess with 3 different types of processes:
 - **Browser** process manages user interface, disk and network I/O
 - **Renderer** process renders web pages, deals with HTML, Javascript. A new renderer created for each website opened
 - 4 Runs in **sandbox** restricting disk and network I/O, minimizing effect of security exploits
 - **Plug-in** process for each type of plug-in



Each tab represents a separate process.





Interprocess Communication

- Processes within a system may be ***independent*** or ***cooperating***
- Cooperating process can affect or be affected by other processes, including sharing data
- Reasons for cooperating processes:
 - Information sharing
 - Computation speedup
 - Modularity
 - Convenience
- Cooperating processes need **interprocess communication (IPC)**
- Two models of IPC
 - **Shared memory**
 - **Message passing**





■ Information Sharing & Collaboration:

- Allows multiple processes to access the same data concurrently (e.g., a database shared by several applications).
- Enables **client-server models** where one process provides services to others.

■ Modularity & Specialization:

- Breaks complex tasks into smaller, independent processes.
- Each process can be developed and maintained separately but work together seamlessly.

■ Data Isolation (Security):

- Processes remain in their own memory spaces (unlike threads).
- IPC acts as a **controlled gateway**—processes only share what they explicitly allow, preventing accidental corruption.





■ **Speedup (Parallelism):**

- Enables **parallel processing** on multi-core systems.
- Example: A video encoder can split frames across 4 processes/cores simultaneously.

■ **Asynchronous Communication:**

- Processes don't have to wait for each other.
- A process can send a message and continue working (non-blocking), improving overall throughput.

■ **Resource Sharing:**

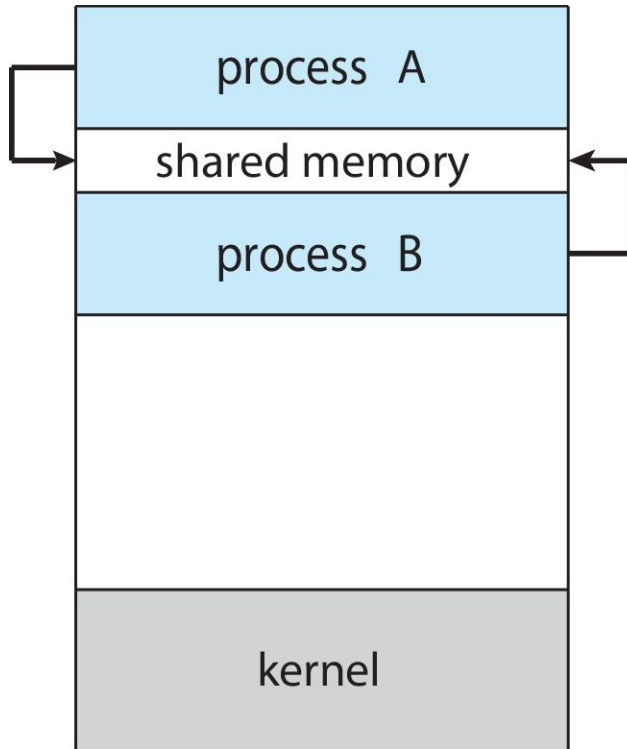
- Avoids duplication.
- Instead of each process loading the same large library or data file into memory, they can share a single copy via IPC (shared memory).





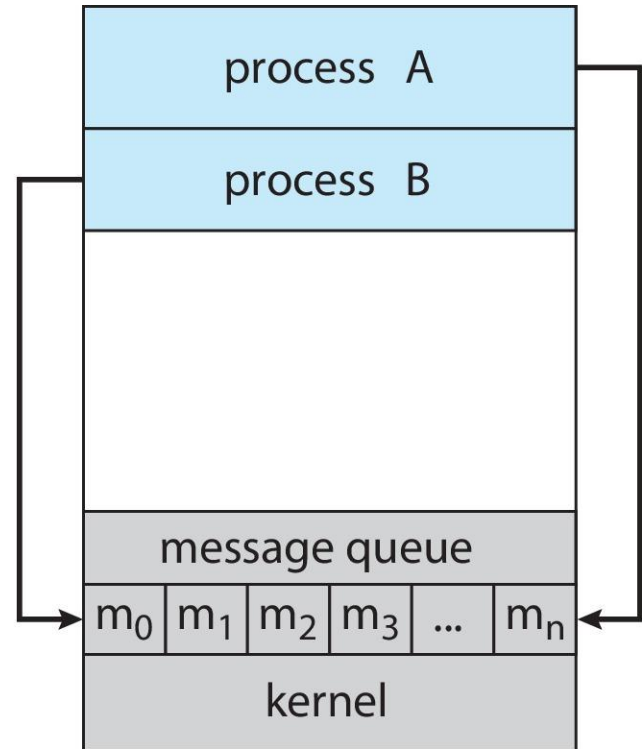
Communications Models

(a) Shared memory.



(a)

(b) Message passing.



(b)





IPC – Shared Memory

■ How it works:

- A region of memory is **shared** between processes (both can read/write to it).
- Processes communicate by directly reading/writing to this shared region.
- **No kernel involvement** after the memory is set up (fast).

■ Key Challenge:

- Requires **synchronization** (locks/semaphores) to prevent race conditions (e.g., two processes writing at the same time).

- **Analogy:** A bulletin board in a public hallway. Anyone can post or read notes, but you need rules so people don't tear notes while others are reading.





```
int main()
{
/* the size (in bytes) of shared memory object */
const int SIZE = 4096;
/* name of the shared memory object */
const char *name = "OS";
/* strings written to shared memory */
const char *message_0 = "Hello";
const char *message_1 = "World!";

/* shared memory file descriptor */
int fd;
/* pointer to shared memory object */
char *ptr;

/* create the shared memory object */
fd = shm_open(name,O_CREAT | O_RDWR,0666);

/* configure the size of the shared memory object */
ftruncate(fd, SIZE);

/* memory map the shared memory object */
ptr = (char *)
    mmap(0, SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0)

/* write to the shared memory object */
sprintf(ptr,"%s",message_0);
ptr += strlen(message_0);
sprintf(ptr,"%s",message_1);
ptr += strlen(message_1);

return 0;
}
```





PIPES

- Pipes are a method of communication between processes in an operating system. They allow one process to send data to another in a unidirectional (one-way) stream.
- A **pipe** is a communication channel that connects:
 - A **writing process** (producer)
 - A **reading process** (consumer)
- Data flows in a **FIFO (First In, First Out)** manner — just like water flowing through a pipe.





Type of Pipes

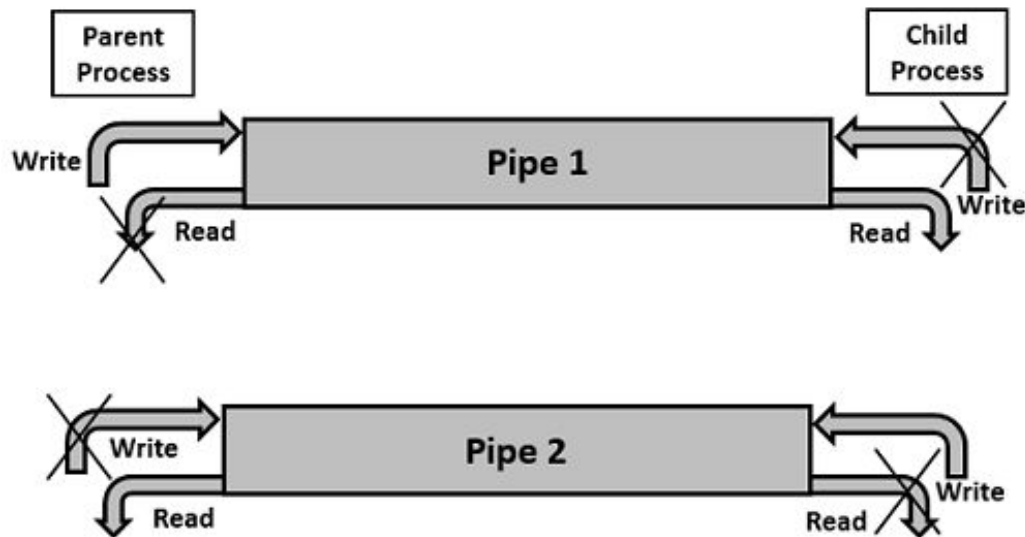
- Unnamed/Anonymous/Ordinary pipes
- Named Pipes





Unnamed Pipes

- Used between parent and child processes
- Created using pipe() system call (in Unix/Linux)
- Exists only while processes are running
- One-way communication
- Cannot be used between unrelated processes





Unnamed/Ordinary Pipe Example

```
int main() {
    int fd[2]; // File descriptors: fd[0] for reading,
               // fd[1] for writing

    pid_t pid;
    char write_msg[] = "Hello from parent!";
    char read_msg[100];

    // Create the pipe
    if (pipe(fd) == -1) {
        perror("pipe");
        exit(EXIT_FAILURE);
    }

    // Fork a child process
    pid = fork();
    if (pid < 0) {
        perror("fork");
        exit(EXIT_FAILURE);
    }

    if (pid == 0) {
        // Child process: read from the pipe
        close(fd[1]); // Close unused write end

        // Read from the pipe
        read(fd[0], read_msg, sizeof(read_msg));
        printf("Child received: %s\n", read_msg);

        close(fd[0]); // Close read end when done
    } else {
        // Parent process: write to the pipe
        close(fd[0]); // Close unused read end

        // Write message to the pipe
        write(fd[1], write_msg, strlen(write_msg) + 1); // +1
        // to include the null terminator

        close(fd[1]); // Close write end when
        // done
        wait(NULL); // Wait for child to finish
    }

    return 0;
}
```





Producer-Consumer Problem

- Paradigm for cooperating processes:
 - *producer* process produces information that is consumed by a *consumer* process
- Two variations:
 - **unbounded-buffer** places no practical limit on the size of the buffer:
 - 4 Producer never waits
 - 4 Consumer waits if there is no buffer to consume
 - **bounded-buffer** assumes that there is a fixed buffer size
 - 4 Producer must wait if all buffers are full
 - 4 Consumer waits if there is no buffer to consume





Bounded-Buffer – Shared-Memory Solution

- Shared data

```
#define BUFFER_SIZE 10
typedef struct {
    . . .
} item;

item buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
```

- Solution is correct, but can only use **BUFFER_SIZE-1** elements





Producer Process – Shared Memory

```
item next_produced;

while (true) {
    /* produce an item in next produced */
    while (((in + 1) % BUFFER_SIZE) == out)
        ; /* do nothing */
    buffer[in] = next_produced;
    in = (in + 1) % BUFFER_SIZE;
}
```





Consumer Process – Shared Memory

```
item next_consumed;

while (true) {
    while (in == out)
        ; /* do nothing */
    next_consumed = buffer[out];
    out = (out + 1) % BUFFER_SIZE;

    /* consume the item in next consumed */
}
```





What about Filling all the Buffers?

- Suppose that we wanted to provide a solution to the consumer-producer problem that fills **all** the buffers.
- We can do so by having an integer **counter** that keeps track of the number of full buffers.
- Initially, **counter** is set to 0.
- The integer **counter** is incremented by the producer after it produces a new buffer.
- The integer **counter** is and is decremented by the consumer after it consumes a buffer.





Producer

```
while (true) {  
    /* produce an item in next produced */  
  
    while (counter == BUFFER_SIZE)  
        ; /* do nothing */  
    buffer[in] = next_produced;  
    in = (in + 1) % BUFFER_SIZE;  
    counter++;  
}
```





Consumer

```
while (true) {  
    while (counter == 0)  
        ; /* do nothing */  
    next_consumed = buffer[out];  
    out = (out + 1) % BUFFER_SIZE;  
    counter--;  
    /* consume the item in next consumed */  
}
```





Race Condition

- **counter++** could be implemented as

```
register1 = counter
register1 = register1 + 1
counter = register1
```

- **counter--** could be implemented as

```
register2 = counter
register2 = register2 - 1
counter = register2
```

- Consider this execution interleaving with “count = 5” initially:

S0: producer execute	register1 = counter	{register1 = 5}
S1: producer execute	register1 = register1 + 1	{register1 = 6}
S2: consumer execute	register2 = counter	{register2 = 5}
S3: consumer execute	register2 = register2 - 1	{register2 = 4}
S4: producer execute	counter = register1	{counter = 6}
S5: consumer execute	counter = register2	{counter = 4}





Race Condition (Cont.)

- Question – why was there no race condition in the first solution (where at most $N - 1$ buffers can be filled?)
- More in Chapter 6.





IPC – Message Passing

- Processes communicate with each other without resorting to shared variables
- IPC facility provides two operations:
 - **send**(*message*)
 - **receive**(*message*)
- The *message* size is either fixed or variable





Message Passing (Cont.)

- If processes P and Q wish to communicate, they need to:
 - Establish a **communication link** between them
 - Exchange messages via send/receive
- Implementation issues:
 - How are links established?
 - Can a link be associated with more than two processes?
 - How many links can there be between every pair of communicating processes?
 - What is the capacity of a link?
 - Is the size of a message that the link can accommodate fixed or variable?
 - Is a link unidirectional or bi-directional?





Implementation of Communication Link

- Physical:
 - Shared memory
 - Hardware bus
 - Network
- Logical:
 - Direct or indirect
 - Synchronous or asynchronous
 - Automatic or explicit buffering





Direct Communication

- Processes must name each other explicitly:
 - **send** (P , *message*) – send a message to process P
 - **receive**(Q , *message*) – receive a message from process Q
- Properties of communication link
 - Links are established automatically
 - A link is associated with exactly one pair of communicating processes
 - Between each pair there exists exactly one link
 - The link may be unidirectional, but is usually bi-directional





Indirect Communication

- Messages are directed and received from mailboxes (also referred to as ports)
 - Each mailbox has a unique id
 - Processes can communicate only if they share a mailbox
- Properties of communication link
 - Link established only if processes share a common mailbox
 - A link may be associated with many processes
 - Each pair of processes may share several communication links
 - Link may be unidirectional or bi-directional





Indirect Communication (Cont.)

- Operations
 - Create a new mailbox (port)
 - Send and receive messages through mailbox
 - Delete a mailbox
- Primitives are defined as:
 - **send**(*A, message*) – send a message to mailbox A
 - **receive**(*A, message*) – receive a message from mailbox A





Indirect Communication (Cont.)

- Mailbox sharing
 - P_1 , P_2 , and P_3 share mailbox A
 - P_1 , sends; P_2 and P_3 receive
 - Who gets the message?
- Solutions
 - Allow a link to be associated with at most two processes
 - Allow only one process at a time to execute a receive operation
 - Allow the system to select arbitrarily the receiver. Sender is notified who the receiver was.

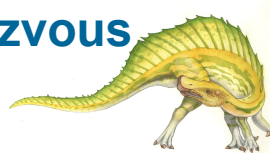




Synchronization

Message passing may be either blocking or non-blocking

- **Blocking** is considered **synchronous**
 - **Blocking send** -- the sender is blocked until the message is received
 - **Blocking receive** -- the receiver is blocked until a message is available
- **Non-blocking** is considered **asynchronous**
 - **Non-blocking send** -- the sender sends the message and continue
 - **Non-blocking receive** -- the receiver receives:
 - 4 A valid message, or
 - 4 Null message
- Different combinations possible
 - If both send and receive are blocking, we have a **rendezvous**





Producer-Consumer: Message Passing

- Producer

```
message next_produced;  
while (true) {  
/* produce an item in next_produced */  
  
    send(next_produced) ;  
}
```

- Consumer

```
message next_consumed;  
while (true) {  
    receive(next_consumed)  
  
/* consume the item in next_consumed */  
}
```





Buffering

- Queue of messages attached to the link.
- Implemented in one of three ways
 1. Zero capacity – no messages are queued on a link.
Sender must wait for receiver (rendezvous)
 2. Bounded capacity – finite length of n messages
Sender must wait if link full
 3. Unbounded capacity – infinite length
Sender never waits





Examples of IPC Systems - POSIX

■ POSIX Shared Memory

- Process first creates shared memory segment

```
shm_fd = shm_open(name, O_CREAT | O_RDWR, 0666);
```

- Also used to open an existing segment
- Set the size of the object

```
ftruncate(shm_fd, 4096);
```

- Use **mmap()** to memory-map a file pointer to the shared memory object
- Reading and writing to shared memory is done by using the pointer returned by **mmap()**.





IPC POSIX Producer

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

int main()
{
    /* the size (in bytes) of shared memory object */
    const int SIZE = 4096;
    /* name of the shared memory object */
    const char *name = "OS";
    /* strings written to shared memory */
    const char *message_0 = "Hello";
    const char *message_1 = "World!";

    /* shared memory file descriptor */
    int shm_fd;
    /* pointer to shared memory object */
    void *ptr;

    /* create the shared memory object */
    shm_fd = shm_open(name, O_CREAT | O_RDWR, 0666);

    /* configure the size of the shared memory object */
    ftruncate(shm_fd, SIZE);

    /* memory map the shared memory object */
    ptr = mmap(0, SIZE, PROT_WRITE, MAP_SHARED, shm_fd, 0);

    /* write to the shared memory object */
    sprintf(ptr, "%s", message_0);
    ptr += strlen(message_0);
    sprintf(ptr, "%s", message_1);
    ptr += strlen(message_1);

    return 0;
}
```





IPC POSIX Consumer

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

int main()
{
    /* the size (in bytes) of shared memory object */
    const int SIZE = 4096;
    /* name of the shared memory object */
    const char *name = "OS";
    /* shared memory file descriptor */
    int shm_fd;
    /* pointer to shared memory object */
    void *ptr;

    /* open the shared memory object */
    shm_fd = shm_open(name, O_RDONLY, 0666);

    /* memory map the shared memory object */
    ptr = mmap(0, SIZE, PROT_READ, MAP_SHARED, shm_fd, 0);

    /* read from the shared memory object */
    printf("%s", (char *)ptr);

    /* remove the shared memory object */
    shm_unlink(name);

    return 0;
}
```





Examples of IPC Systems - Mach

- Mach communication is message based
 - Even system calls are messages
 - Each task gets two ports at creation - Kernel and Notify
 - Messages are sent and received using the `mach_msg()` function
 - Ports needed for communication, created via
`mach_port_allocate()`
 - Send and receive are flexible; for example four options if mailbox full:
 - 4 Wait indefinitely
 - 4 Wait at most n milliseconds
 - 4 Return immediately
 - 4 Temporarily cache a message





Mach Messages

```
#include<mach/mach.h>

struct message {
    mach_msg_header_t header;
    int data;
};

mach port t client;
mach port t server;
```





Mach Message Passing - Client

```
/* Client Code */

struct message message;

// construct the header
message.header.msgh_size = sizeof(message);
message.header.msgh_remote_port = server;
message.header.msgh_local_port = client;

// send the message
mach_msg(&message.header, // message header
        MACH_SEND_MSG, // sending a message
        sizeof(message), // size of message sent
        0, // maximum size of received message - unnecessary
        MACH_PORT_NULL, // name of receive port - unnecessary
        MACH_MSG_TIMEOUT_NONE, // no time outs
        MACH_PORT_NULL // no notify port
    );
```





Mach Message Passing - Server

```
/* Server Code */

struct message message;

// receive the message
mach_msg(&message.header, // message header
        MACH_RCV_MSG, // sending a message
        0, // size of message sent
        sizeof(message), // maximum size of received message
        server, // name of receive port
        MACH_MSG_TIMEOUT_NONE, // no time outs
        MACH_PORT_NULL // no notify port
    );
```





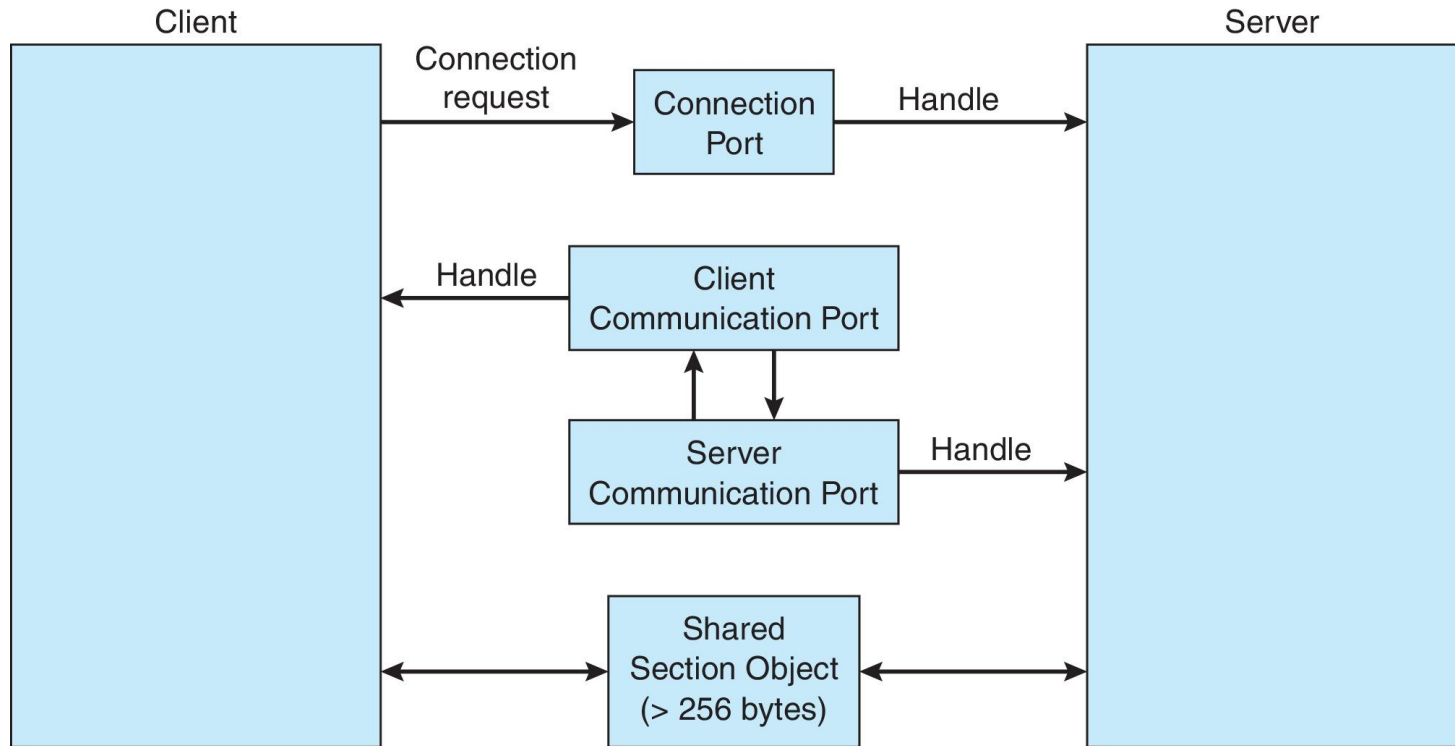
Examples of IPC Systems – Windows

- Message-passing centric via **advanced local procedure call (LPC)** facility
 - Only works between processes on the same system
 - Uses ports (like mailboxes) to establish and maintain communication channels
 - Communication works as follows:
 - 4 The client opens a handle to the subsystem's **connection port** object.
 - 4 The client sends a connection request.
 - 4 The server creates two private **communication ports** and returns the handle to one of them to the client.
 - 4 The client and server use the corresponding port handle to send messages or callbacks and to listen for replies.





Local Procedure Calls in Windows





Pipes

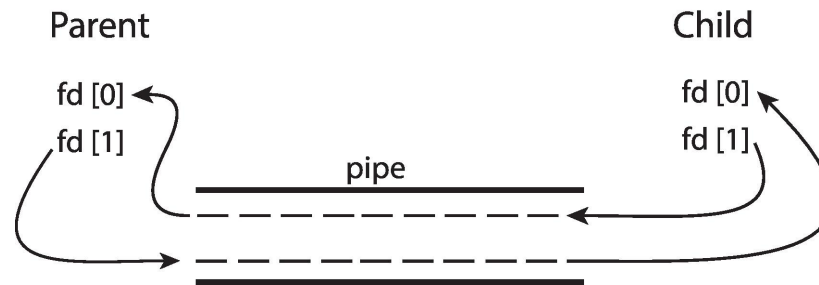
- Acts as a conduit allowing two processes to communicate
- Issues:
 - Is communication unidirectional or bidirectional?
 - In the case of two-way communication, is it half or full-duplex?
 - Must there exist a relationship (i.e., **parent-child**) between the communicating processes?
 - Can the pipes be used over a network?
- **Ordinary pipes** – cannot be accessed from outside the process that created it. Typically, a parent process creates a pipe and uses it to communicate with a child process that it created.
- **Named pipes** – can be accessed without a parent-child relationship.





Ordinary Pipes

- Ordinary Pipes allow communication in standard producer-consumer style
- Producer writes to one end (the **write-end** of the pipe)
- Consumer reads from the other end (the **read-end** of the pipe)
- Ordinary pipes are therefore unidirectional
- Require parent-child relationship between communicating processes



- Windows calls these **anonymous pipes**





Named Pipes

- Named Pipes are more powerful than ordinary pipes
- Communication is bidirectional
- No parent-child relationship is necessary between the communicating processes
- Several processes can use the named pipe for communication
- Provided on both UNIX and Windows systems





Communications in Client-Server Systems

- Sockets
- Remote Procedure Calls





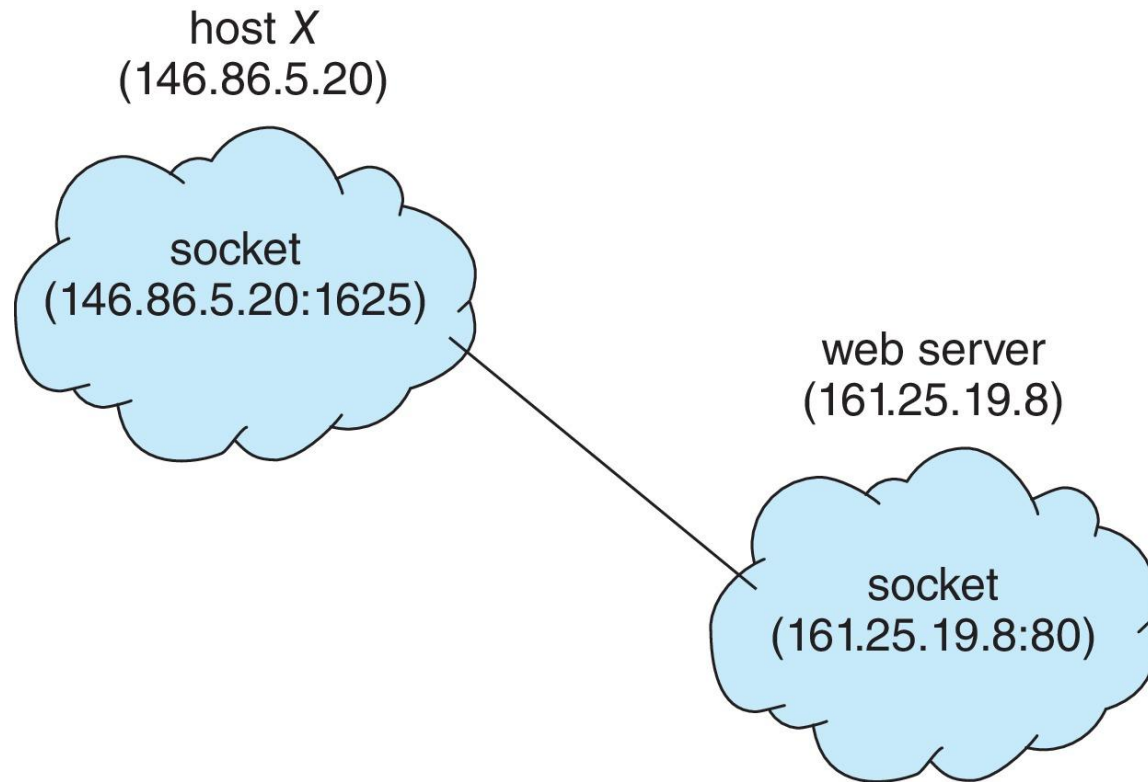
Sockets

- A **socket** is defined as an endpoint for communication
- Concatenation of IP address and **port** – a number included at start of message packet to differentiate network services on a host
- The socket **161.25.19.8:1625** refers to port **1625** on host **161.25.19.8**
- Communication consists between a pair of sockets
- All ports below 1024 are **well known**, used for standard services
- Special IP address 127.0.0.1 (**loopback**) to refer to system on which process is running





Socket Communication





Sockets in Java

- Three types of sockets
 - **Connection-oriented (TCP)**
 - **Connectionless (UDP)**
 - **MulticastSocket** class– data can be sent to multiple recipients
- Consider this “Date” server in Java:

```
import java.net.*;
import java.io.*;

public class DateServer
{
    public static void main(String[] args) {
        try {
            ServerSocket sock = new ServerSocket(6013);

            /* now listen for connections */
            while (true) {
                Socket client = sock.accept();

                PrintWriter pout = new
                    PrintWriter(client.getOutputStream(), true);

                /* write the Date to the socket */
                pout.println(new java.util.Date().toString());

                /* close the socket and resume */
                /* listening for connections */
                client.close();
            }
        }
        catch (IOException ioe) {
            System.err.println(ioe);
        }
    }
}
```





Sockets in Java

The equivalent Date client

```
import java.net.*;
import java.io.*;

public class DateClient
{
    public static void main(String[] args) {
        try {
            /* make connection to server socket */
            Socket sock = new Socket("127.0.0.1",6013);

            InputStream in = sock.getInputStream();
            BufferedReader bin = new
                BufferedReader(new InputStreamReader(in));

            /* read the date from the socket */
            String line;
            while ( (line = bin.readLine()) != null)
                System.out.println(line);

            /* close the socket connection*/
            sock.close();
        }
        catch (IOException ioe) {
            System.err.println(ioe);
        }
    }
}
```





Remote Procedure Calls

- Remote procedure call (RPC) abstracts procedure calls between processes on networked systems
 - Again uses ports for service differentiation
- **Stubs** – client-side proxy for the actual procedure on the server
- The client-side stub locates the server and **marshalls** the parameters
- The server-side stub receives this message, unpacks the marshalled parameters, and performs the procedure on the server
- On Windows, stub code compile from specification written in **Microsoft Interface Definition Language (MIDL)**





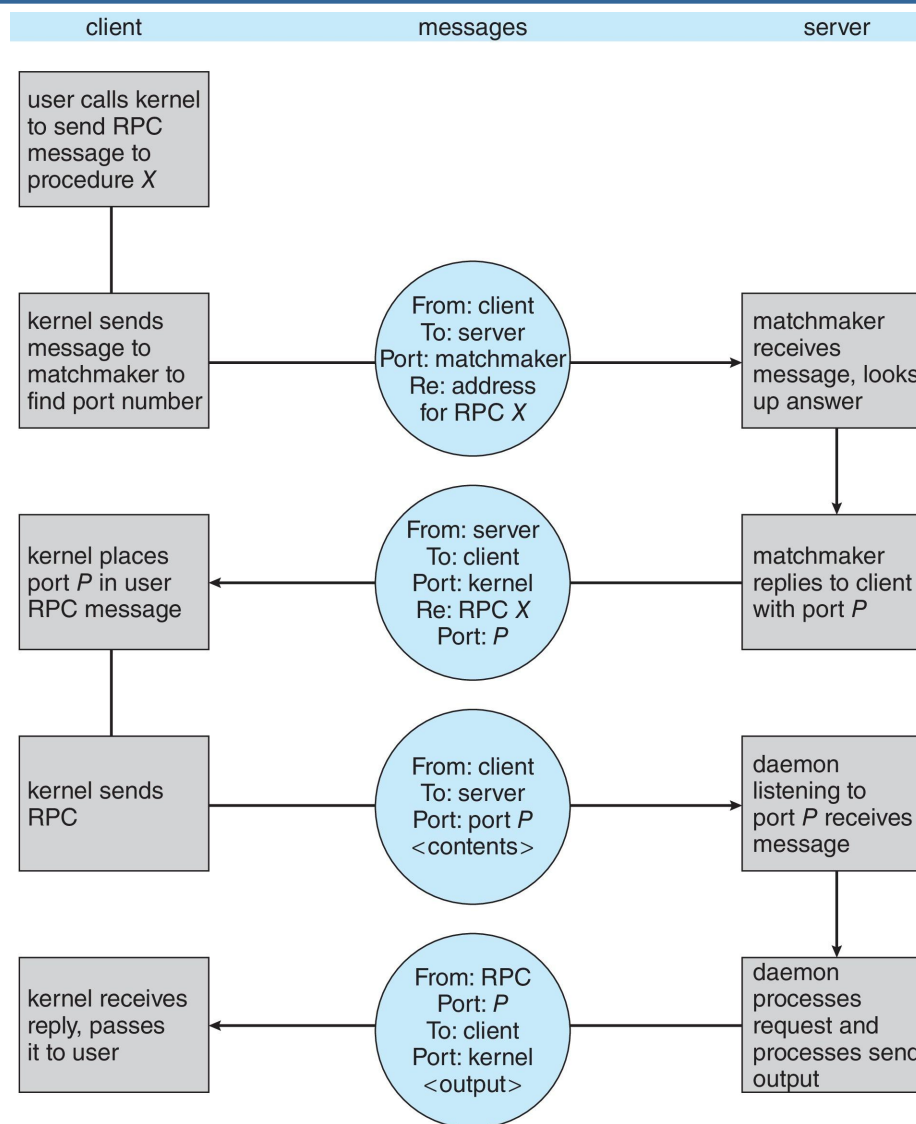
Remote Procedure Calls (Cont.)

- Data representation handled via **External Data Representation (XDL)** format to account for different architectures
 - **Big-endian** and **little-endian**
- Remote communication has more failure scenarios than local
 - Messages can be delivered ***exactly once*** rather than ***at most once***
- OS typically provides a rendezvous (or **matchmaker**) service to connect client and server





Execution of RPC



End of Chapter 3

